Markets From Zero

How the finance industry actually works: prices, rules, players, math. A self-study roadmap with milestones so progress (or lack of it) is visible.

How this works

- **The loop for every phase:** read, then close the book and answer the gate questions in writing, then score yourself, then re-study whatever scored low, then do the milestone project.
- **The scoring rubric (per question):** 0 = cannot answer. 1 = vague or partly wrong. 2 = correct but needed notes. 3 = correct from memory, in plain words, with an example.
- **Pass a phase** when your average is 2.5 or higher AND the milestone is done. Not before. Moving on with gaps is how you end up with fake understanding.
- **Fleshing out:** at the end of each phase, run a gap hunt. Write down 5 terms from the reading you could not define without looking, and look them up. Then bring your answers here and I will quiz you and push on the weak spots, one question at a time.
- **Spaced re-test:** at the end of every phase, re-answer 5 random gate questions from earlier phases. If an old answer has decayed, that is a real signal, not a failure.

Time budget

Phase	Topic	Hours	Weeks at 5 hrs/wk
0	Orientation and vocabulary	5	1
1	Accounting and financial statements	10	2
2	Valuation and corporate finance	25	5
3	Market structure and the rules	25	5
4	Macro, rates, and bonds	25	5
5	Derivatives and options	30	6
6	Portfolio theory and risk	15	3
7	Quant toolkit (probability, stats, Python)	50	10
8	Case studies (runs in parallel)	ongoing	-
Total		**~185**	**~37 (about 9 months)**

Hour estimates are rough guesses. Track your real numbers in the weekly log and adjust. At 8 hrs/wk the whole thing is closer to 6 months.

A note on sourcing books

I will not point you to pirated copies, but you do not need them. Borrow nearly all of these free and legally through your library's Libby or Hoopla apps, Open Library and the Internet Archive lending library, or an interlibrary loan request. Several resources below are free outright: Damodaran's lectures and data, the SEC guides, the ISLR textbook, FRED, and Stat 110. Older editions of Hull, Harris, and the corporate finance texts cover the same ground and are cheap or easy to borrow.

Phase 0: Orientation (do this first, in week one)

Budget: 5 hours | **Goal:** stop being lost when people talk. Build the vocabulary and set up the free information diet that you keep for the whole journey.

Do this

- Subscribe to **Matt Levine's Money Stuff** (free Bloomberg newsletter). It is the best plain-English window into how deals, markets, and rules actually work, with jokes. Read it daily for the whole roadmap, even when you only half understand it.
- Listen to **Odd Lots** (Bloomberg podcast). Pick episodes on topics you are currently studying.
- Watch **Ray Dalio, How the Economic Machine Works** (about 30 min, free on YouTube) for a rough map of credit, debt, and cycles. Treat it as an intro, not gospel.
- Start a **glossary file**. Every term you hit and cannot define goes in. Define it in your own words, once, and never copy a dictionary line.

Gate questions

- In one paragraph each: what is a stock, a bond, and an index fund? What does each owner actually own or get paid?
- What is the difference between a broker, a market maker, and an exchange?

MILESTONE	Glossary has 30+ terms in your own words. You have read at least 5 Money Stuff issues and can summarize one in 3 sentences.

Phase 1: Accounting and financial statements

Budget: 10 hours | **Goal:** read a company's financials and know whether it is healthy, how it makes money, and where the bodies are buried.

Resources (read selectively, you do not need every chapter)

- *Financial Statements* by Thomas Ittelson (short, practical, the best first book).
- SEC's free *Beginners' Guide to Financial Statements* (sec.gov).
- Khan Academy, Finance and Capital Markets, accounting basics (free videos if reading is slow).
- Practice document: a real 10-K from SEC EDGAR (see the niche section).

Rules and knowledge you must be able to state

- The three statements (income, balance sheet, cash flow) and exactly how they link.
- Accrual vs cash accounting. Revenue recognition. Working capital (inventory, receivables, payables).
- Net income vs EBITDA vs free cash flow, and why managers like to quote the flattering one.
- Leverage: debt, interest coverage, debt maturities. GAAP vs non-GAAP numbers.

Gate questions: answer in writing, from memory, in plain words

- Why can a profitable company go bankrupt?
- A company sells $100 on credit. Walk through what changes on all three statements.
- Why does AutoZone's balance sheet show negative shareholder equity? (Find out. Is it a red flag or a choice?)
- Where in a 10-K do you find debt maturities, risks, and what management admits to worrying about?

MILESTONE	Read the latest AutoZone 10-K (you know the business from the inside, which makes the numbers concrete). Write one page: how it makes money, margins, inventory, debt, one real risk. Compare one number to what you see at work.

Phase 2: Valuation and corporate finance

Budget: 25 hours | **Goal:** understand what a stock is worth in theory, and why the market price differs.

Resources (read selectively, you do not need every chapter)

- Aswath Damodaran (NYU): free Investment Valuation lectures on YouTube and his site. Best teacher on the topic, period.
- *The Little Book of Valuation* (Damodaran) as the short companion; his full *Investment Valuation* as the reference.
- *Corporate Finance* by Berk and DeMarzo (time value of money, NPV, CAPM, WACC chapters).
- *Valuation* by McKinsey/Koller for deeper practice once the basics click.

Rules and knowledge you must be able to state

- Time value of money, discounting, NPV. Risk vs return, CAPM, beta, cost of capital (WACC).
- Discounted cash flow: forecasts, terminal value, why terminal value dominates the answer.
- Multiples (P/E, EV/EBITDA) and what growth and risk they imply.
- Equity risk premium, and how interest rates feed into every valuation.

Gate questions: answer in writing, from memory, in plain words

- Why does a rate hike hit growth stocks harder than boring ones?
- A stock trades at a P/E of 40. What does the market have to believe about growth?
- Why is a DCF so sensitive to its assumptions, and what does that say about confident price targets?
- What does 'priced in' mean in numbers, not words?

MILESTONE	Build a DCF spreadsheet for one company (AutoZone or another you know). Make three scenarios (bear, base, bull). Write half a page: what the market price implies, and which assumption you think the market is wrong about, if any.

Phase 3: Market structure and the rules of the system

Budget: 25 hours | **Goal:** know how trading actually works mechanically, and the laws and plumbing that constrain everyone.

Resources (read selectively, you do not need every chapter)

- *Trading and Exchanges* by Larry Harris. The classic. Dense, so skim for concepts, read the order-types and liquidity chapters carefully.
- *Empirical Market Microstructure* by Joel Hasbrouck (short, more math).
- FINRA's SIE exam content outline (free at finra.org). Use it as a checklist of the rules a real industry entry-level person must know. You do not have to take the exam.
- SEC.gov explainers on Regulation NMS, Regulation SHO, and market-wide circuit breakers. *Flash Boys* (Lewis) as narrative, but it is contested, so cross-check claims.

Rules and knowledge you must be able to state

- Order types: market, limit, stop, IOC, FOK. Bid, ask, spread, depth, the order book, and the NBBO.
- Market makers, maker-taker fees, payment for order flow, dark pools, opening and closing auctions.
- Clearing and settlement: central counterparties, DTCC, and the US move to T+1 settlement (verify current).
- Short selling (locates, Reg SHO), margin (Reg T), the pattern day trader rule, circuit breakers and limit up/limit down, trading halts.
- Insider trading and disclosure: Rule 10b-5, Reg FD, Form 4, 13F, 13D, wash-sale tax rule.
- **Verify every rule against the SEC or FINRA site before relying on it.** Thresholds and rules change.

Gate questions: answer in writing, from memory, in plain words

- Who is on the other side when you click buy, and how do they make money? When do they lose?
- Why is the spread wider on a thin stock and during volatility?
- What happens, step by step, between clicking buy and legally owning the shares?
- Mechanically, what is a short squeeze? Why is the downside of shorting unlimited?
- Why can brokers legally restrict trading in a stock during a frenzy?

MILESTONE	Open a free paper trading account (Interactive Brokers paper or thinkorswim paperMoney). Place limit and market orders on a liquid ETF and a thinly traded stock; record the spread you paid on each. Then write a one-page mechanical explanation of the January 2021 GameStop episode (short interest, gamma, clearing collateral).

Phase 4: Macro, rates, and bonds

Budget: 25 hours | **Goal:** understand why interest rates are the gravity of all financial markets.

Resources (read selectively, you do not need every chapter)

- Free: FRED (St. Louis Fed data), Fed FOMC statements and minutes, and Dalio's video from Phase 0.
- *The Economics of Money, Banking and Financial Markets* by Mishkin for the central bank and banking machinery.
- *Fixed Income Securities* by Tuckman (rigorous) or Fabozzi's bond text (more practical) for bond math.
- Optional: Odd Lots episodes on the repo market and Treasury market plumbing.

Rules and knowledge you must be able to state

- How money is created (bank lending, reserves), what the policy rate is, QE and QT.
- Inflation measures (CPI vs PCE) and how the Fed's dual mandate shapes decisions.
- Bonds: price vs yield, duration, convexity, credit spreads, the yield curve and what it signals.
- The repo market, the dollar, and how rates reach stocks, housing, and credit.

Gate questions: answer in writing, from memory, in plain words

- Why does a bond's price fall when yields rise? Do the math with a simple 1-year example.
- What is an inverted yield curve supposed to signal, and what are the caveats?
- What is the Fed's actual tool for hitting a rate target?
- Why do bond yields matter to stock valuations (link to Phase 2)?

MILESTONE	Before the next FOMC decision: write your prediction of the decision and of how stocks and the 2-year yield react, with probabilities. After: score yourself and write why you were right or wrong.

Phase 5: Derivatives and options

Budget: 30 hours | **Goal:** know how options, futures and swaps work and are priced, and why volatility is a tradable thing.

Resources (read selectively, you do not need every chapter)

- *Options, Futures, and Other Derivatives* by John Hull. The standard text. Focus on the options chapters, Greeks, and Black-Scholes.
- *Option Volatility and Pricing* by Sheldon Natenberg (practical, trader-oriented).
- Cboe's free education pages for contract specs and the VIX.
- Later, optional: *Dynamic Hedging* (Taleb) and *Volatility Trading* (Sinclair).

Rules and knowledge you must be able to state

- Forwards and futures: margin and daily mark-to-market. Hedging vs speculation.
- Options: calls, puts, payoff diagrams, put-call parity, intrinsic vs time value.
- The Greeks (delta, gamma, theta, vega) and delta hedging. Implied vs realized volatility, the volatility smile.
- Swaps and how banks use them. Why leverage cuts both ways.

Gate questions: answer in writing, from memory, in plain words

- What does an option actually give you? What is the maximum loss of the buyer vs the seller?
- Why does an option cost more when volatility goes up?
- State put-call parity in words and explain why it must hold.
- What is a gamma squeeze, mechanically? (Links to Phase 3.)

MILESTONE	Implement Black-Scholes in Python from scratch. Check it against a known example, plot delta and vega against the stock price, then explain in half a page why real option prices show a volatility smile instead of one volatility number.

Phase 6: Portfolio theory and risk

Budget: 15 hours | **Goal:** know how professionals think about risk and returns, and why most lose to the index.

Resources (read selectively, you do not need every chapter)

- *A Random Walk Down Wall Street* by Malkiel (intro).
- *Investments* by Bodie, Kane, Marcus (the standard text; read the portfolio theory and efficient-market chapters).
- *The Most Important Thing* by Howard Marks (how experienced investors think about risk).
- *A Man for All Markets* by Ed Thorp (position sizing, the Kelly idea, real practice).

Rules and knowledge you must be able to state

- Diversification, correlation, Sharpe ratio, beta, drawdown.
- The efficient market hypothesis and the strongest arguments against it; factor investing (value, momentum).
- Position sizing and the Kelly criterion. Value at Risk and why it fails in crises.

Gate questions: answer in writing, from memory, in plain words

- What is the difference between being wrong and being ruined?
- Why do most active managers fail to beat an index fund after fees?
- Why does correlation jump toward 1 during crises, and what does that do to diversification?

MILESTONE	Using free data (FRED or Yahoo Finance), compute the return, Sharpe ratio, and max drawdown for a 60/40 portfolio vs the S&P; 500 over 20+ years. Write what each number teaches and what it hides.

Phase 7: Quant toolkit (probability, statistics, Python)

Budget: 50 hours | **Goal:** get the math and code to test ideas honestly, and to see why most trading edges are fake.

Resources (read selectively, you do not need every chapter)

- Harvard Stat 110 (free lectures) with *Introduction to Probability* by Blitzstein and Hwang.
- 3Blue1Brown's Essence of Linear Algebra (free) for intuition.
- *An Introduction to Statistical Learning* (ISLR): free official PDF from its authors.
- *Python for Data Analysis* by McKinney (pandas).
- *Advances in Financial Machine Learning* by Lopez de Prado (focus on why backtests overfit).
- *Quantitative Trading* by Ernest Chan, and for interview-style problems: *Heard on the Street* (Crack) and *A Practical Guide to Quantitative Finance Interviews* (Zhou).

Rules and knowledge you must be able to state

- Conditional probability, expected value, variance, Bayes, distributions, the central limit theorem.
- Regression, hypothesis tests, p-hacking and multiple-testing, cross-validation.
- Backtesting traps: look-ahead bias, survivorship bias, overfitting, ignoring costs and slippage.
- Clean data handling in pandas; time series basics.

Gate questions: answer in writing, from memory, in plain words

- Why do most backtests look great and then fail live?
- You test 100 strategies and the best has a p-value of 0.01. What should you conclude?
- What costs does a backtest typically leave out, and how big can they be?
- Solve 10 expected-value and probability puzzles from *Heard on the Street* without looking at answers first.

MILESTONE	Build a simple strategy (for example, momentum on ETFs). Test it on one period and validate on a held-out period. Include transaction costs. Write a short post-mortem: why this probably would not survive live trading, and what would have to be true for it to.

Phase 8: Case studies and narrative (run in parallel from Phase 3 onward)

Every blowup is a free lesson in which rule or assumption failed. For each case, write a half page answering: **What was the mechanism? Which assumption broke? Which rule or incentive made it worse? What would have saved them?**

Case	What to learn
LTCM (1998)	Leverage plus correlated bets; models fail in the tails. Book: *When Genius Failed*.
2008 financial crisis	Mortgage securitization, leverage, counterparty risk. Books: *The Big Short*, *Too Big to Fail*.
Flash Crash (2010)	Liquidity vanishing, how one algo can cascade; origin of limit up/limit down.
Knight Capital (2012)	Deployment error in trading software lost about $440M in under an hour. Operational risk.
Volmageddon (Feb 2018)	Short-volatility products and how structure forced buying into a spike.
GameStop (Jan 2021)	Short squeeze, gamma, clearing collateral demands, retail vs institutions.
Archegos (2021)	Total return swaps, hidden leverage, why banks had no idea of the combined size.
Silicon Valley Bank (2023)	Duration mismatch, deposit runs, interest rate risk on 'safe' assets.

Culture and perspective reading (for what the show Industry is showing): *Liar's Poker* (Lewis), *The Man Who Solved the Market* (Zuckerman, on Renaissance), *Fortune's Formula* (Poundstone), *Models. Behaving. Badly.* (Derman). Treat them as stories and verify factual claims.

Niche goldmines (free, and rarely taught to beginners)

- **SEC EDGAR** (sec.gov/edgar): the original documents. 10-K (annual), 10-Q (quarterly), 8-K (events), **Form 4** (insider buying and selling), **13F** (what big funds hold, quarterly), **13D/G** (5%+ stakes). Reading primary filings beats reading headlines.
- **Earnings call transcripts**: the Q&A at the end is where analysts poke at what management dodges. Read the Q&A of three calls for one company and note what gets evaded.
- **Damodaran's data page** (NYU Stern, free): industry multiples, margins, and the implied equity risk premium updated regularly. Great sanity check for your DCFs.
- **FRED** (fred.stlouisfed.org): free economic data you can chart and download. Learn it and you can test any macro claim.
- **Fed FOMC minutes and press conferences**: how to read what changed from one statement to the next is itself a skill.
- **Prediction journal with Brier scores**: record predictions with probabilities, then score them. It exposes overconfidence faster than anything else.
- **Post-mortem habit**: when a stock you follow moves 10% in a day, find the 8-K or news and write why before reading the explanation.
- **Paper trade with rules**: every paper trade needs a written reason, a size, and a stop condition before entry. A journal of reasons is the whole point, not the P&L.
- **Learn the incentives**: ask of every actor (broker, analyst, fund, company, regulator) who pays them and for what. It explains most behavior in the industry.

Progress system

Milestone checklist

Done	Phase	Milestone	Avg score
[ ]	0	Glossary 30+ terms, 5 Money Stuff issues, one 3-sentence summary	-
[ ]	1	AutoZone 10-K one-page analysis; gate avg 2.5+	____
[ ]	2	DCF with 3 scenarios and written comparison to market price	____
[ ]	3	Paper trades with spread recorded; GameStop mechanics page	____
[ ]	4	FOMC prediction made and scored	____
[ ]	5	Black-Scholes in Python with Greek plots and smile explanation	____
[ ]	6	60/40 vs S&P; risk and return write-up	____
[ ]	7	Strategy backtest with held-out test and honest post-mortem	____
[ ]	8	At least 4 case study half-pages written	-

Weekly log (copy this into a notes file each week)

Item	Entry
Week / dates	
Hours studied (real, not planned)	
Pages or lectures completed	
One thing I understand now that I did not last week	
One thing that is still confusing	
Gate questions attempted / avg score	
Plan for next week	

How to tell if you are making progress

- **Real progress:** gate scores rise on re-test, you can explain a news event mechanically, and your written answers get shorter and clearer.
- **No progress:** you re-read without writing, scores stay at 0-1 after re-study, or the log has gaps. That is useful information, not failure.
- **If two weeks go by with no log entry:** halve the weekly target instead of quitting. Five hours a week for months beats a heroic week followed by silence.
- **Skip rule:** if a book is not clicking, switch to the free lectures for that phase and come back to the book later. The goal is the knowledge, not finishing the book.

What this roadmap deliberately leaves out

It is not trading advice and will not teach you to beat the market. Most of the value is understanding how the system works and its rules. Rules, thresholds, and fees change, so confirm anything specific on SEC.gov or FINRA.org before acting on it. Tax rules are their own subject; talk to a tax professional before trading real money.